The Sombrero galaxy (M104, type Sa/b). (The Cerro Tololo Inter-American Observatory.)

THE EXPANDING UNIVERSE & THE BIG BANG MODEL

The Expanding Universe and The Big Bang Model of the orgin of the Universe

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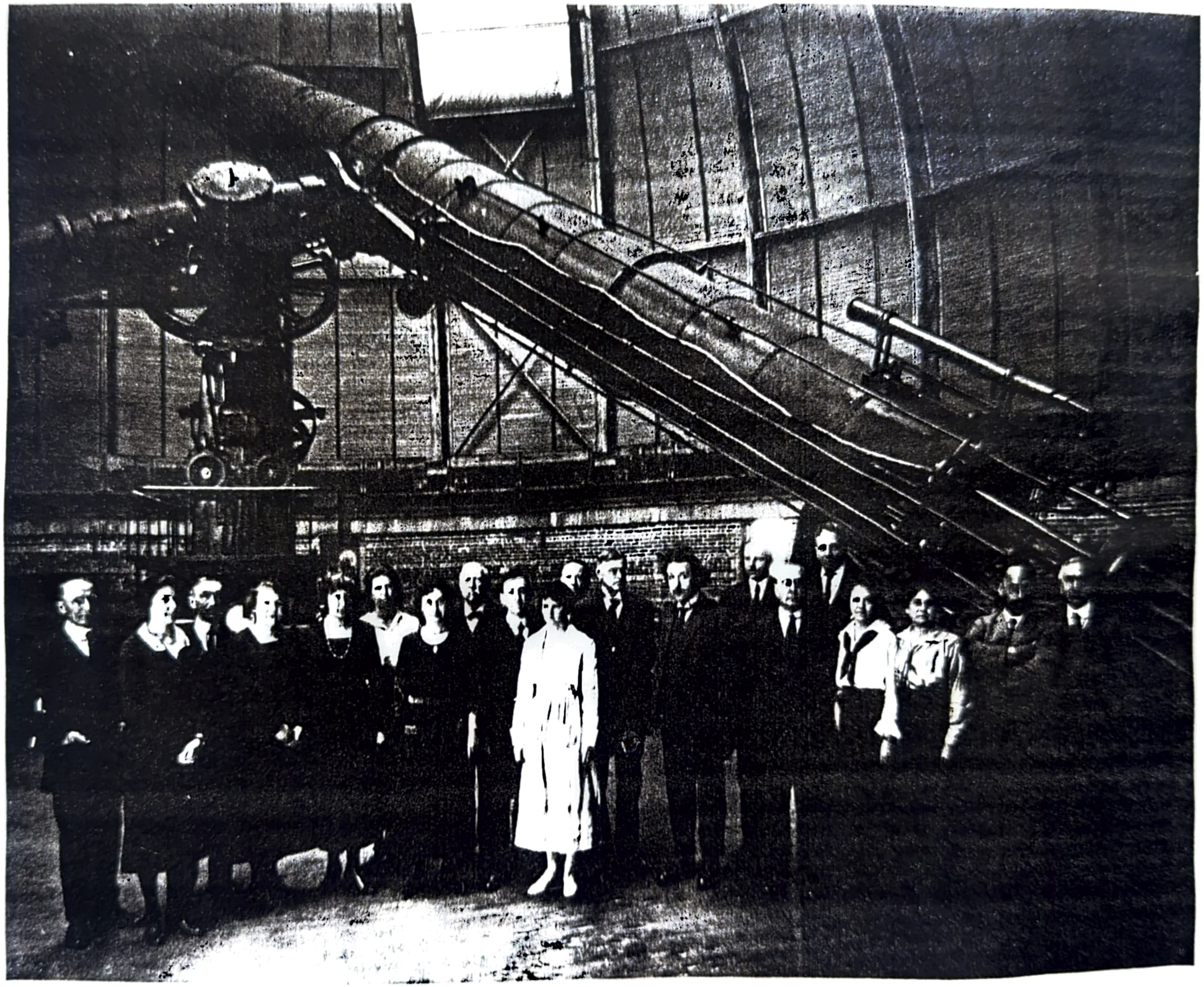


Figure 2.16. In 1921, Albert Einstein posed with the staff of Yerkes Observatory in front of the 40-inch refractor, the largest in the world. (Yerkes Observatory photograph, University of Chicago, Williams Bay, Wisconsin.)

Amidst the boundless, infinite Universe man and his world is but just a mere speck of physical insignificance. But man's imagination is infinite and boundless; and those greats of our race, who have put their imagination, intuition and deduction to harmonious use, have come up with amazing facts of our universe and great theories to account for the observation.

In the following account two of the greatest scientific discoveries of our time, that is, the discovery of the Expanding Universe and the Big Bang Theory of the origin of the Universe are presented with an account of the observational and theoratical methods employed in the discoveries

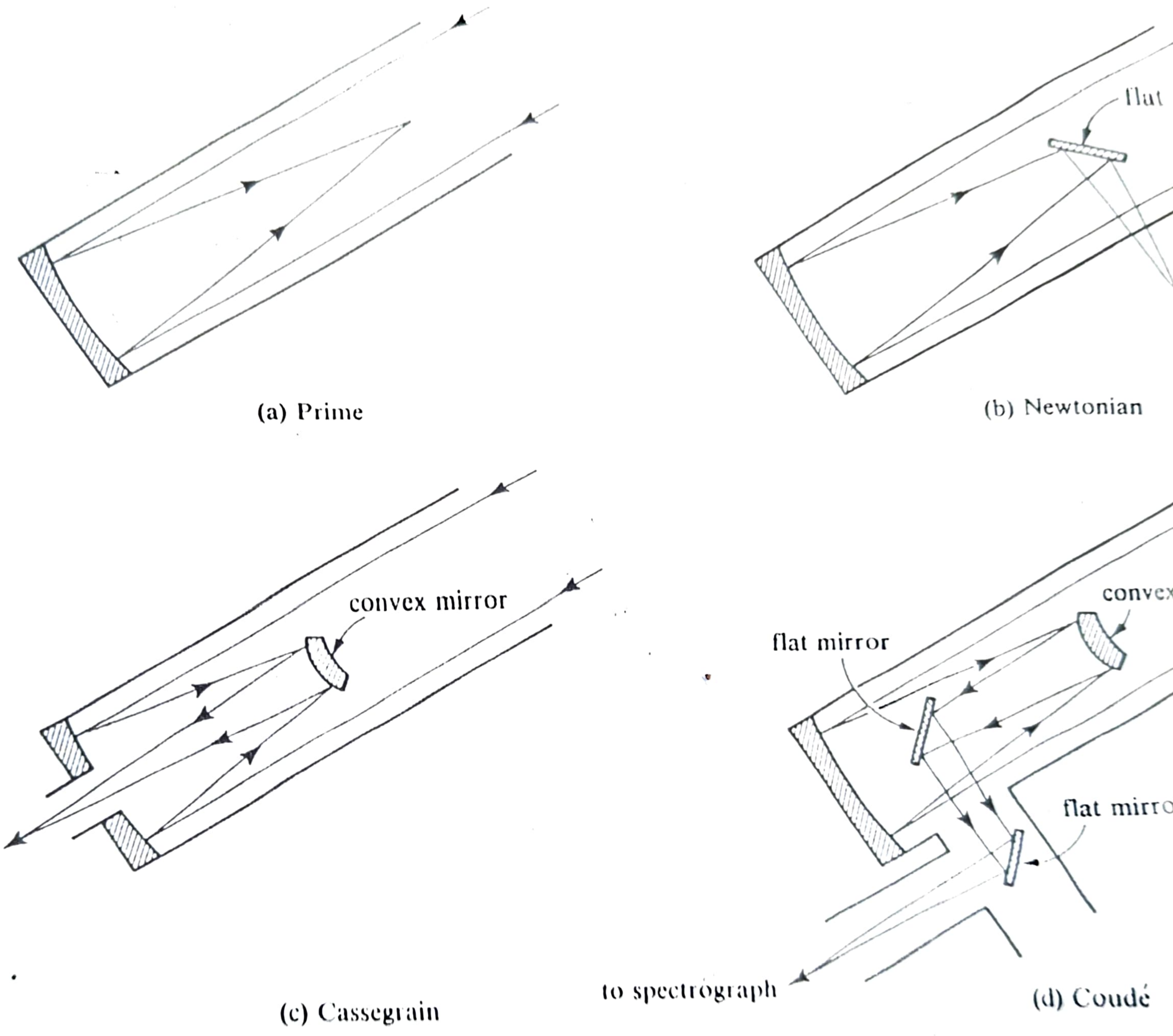


Figure 2.24. Various arrangements for re
the prime focus in front of the primary mi
focus at the side of the telescope; (c) the
back of the telescope; (d) the Coudé foc
scope.

Fig(1)

The Expanding Universe

1 Methods in Observational Astronomy

Astronomical Telescopes

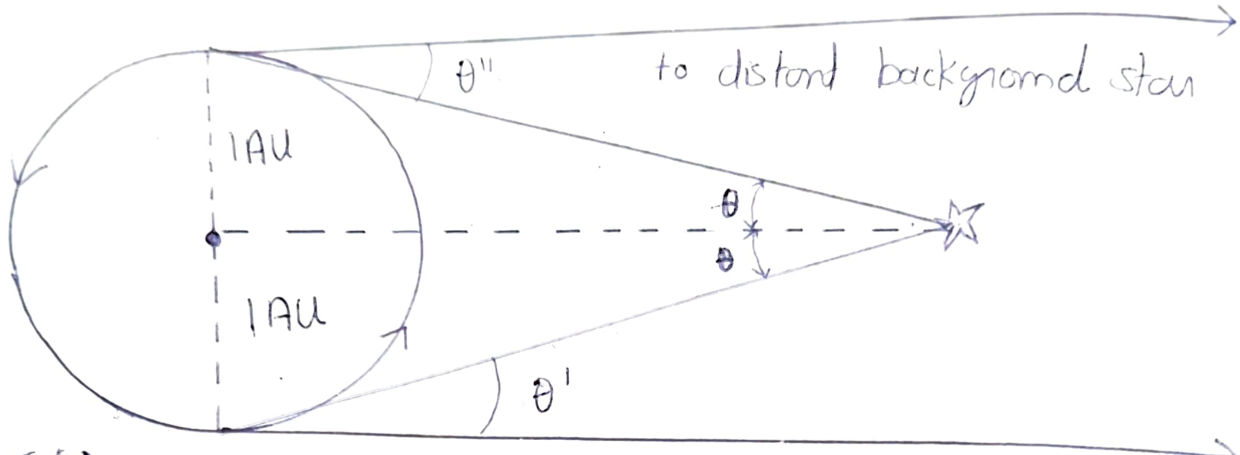
Astronomical telescopes can be constructed which use either refraction or reflection to gather light over a large collecting surface and focus it on a much smaller area. This property gathering more light and not magnification, is the primary purpose of most astronomical telescopes.

Though there are two types of telescopes, refracting and reflecting, the most widely used are the reflecting telescopes. Refracting telescopes are not widely used because of their many drawbacks like chromatic abberation, limitation in the design and setting up of lens systems etc. So we shall see about reflecting telescopes here.

Reflecting Telescopes

The main element in reflecting telescopes is a parabolic mirror which brings all light from a point source to a focus at a single point. Since secondary mirrors can be easily used to deflect a light beam, a variety of possible foci can be

Trignometric Parallax Method



Fig(2)

Distance measurement by trignometric parallax

Approximating the background star to be infinitely far away, it is easy to see that

$$(\theta' + \theta'') = 2\theta$$

If θ is measured in radians, the small-angle formula yeilds the distance to the nearby star as $r = 1AU/\theta$

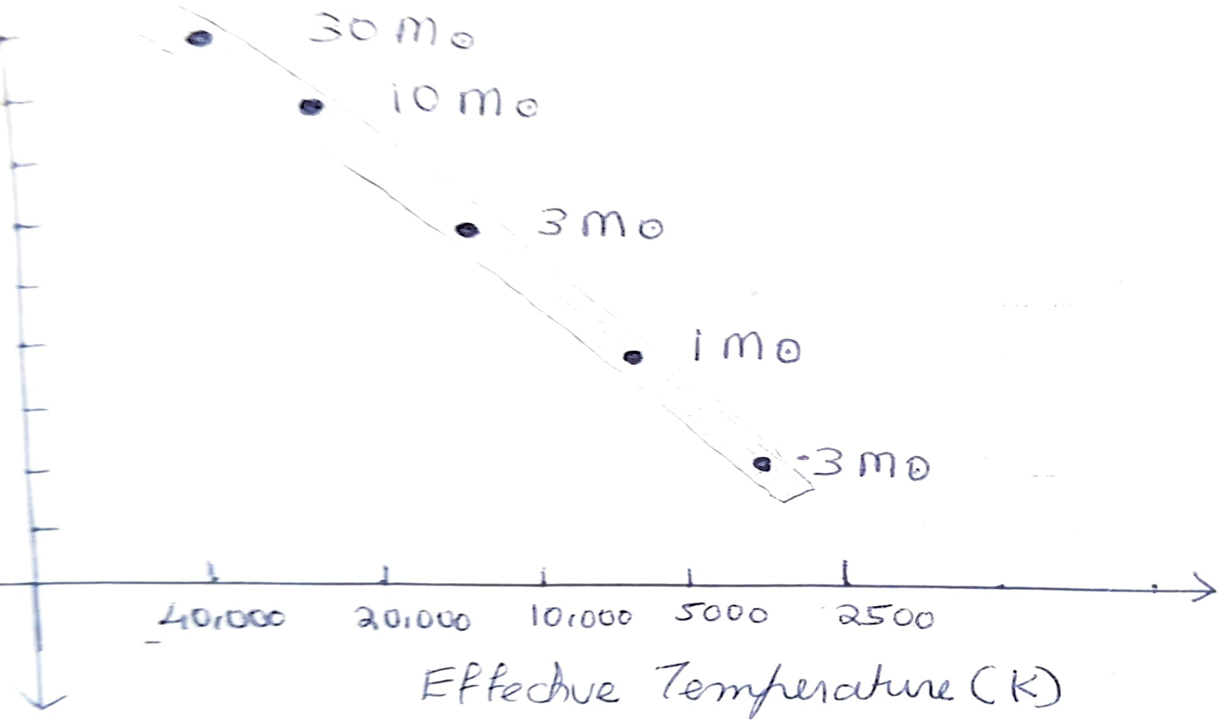
arranged, with different focal ratios, so that a single telescope can be used equally efficiently for several different purposes. The various foci in common use are shown in fig (1)

Estimating the distance scale of the Universe

a Trigonometric parallax method.

This was the earliest method used to gauge the distance of nearby stars. See Fig (2). During six months of the Earths motion about the sun, a nearby star is observed to shift its ~~position~~ apparent angular position with respect to distant background stars. In the example shown in fig (2) it is first observed to lie at an angle θ' to the left of a distant background star; six months later it is seen at an angle θ'' to the right of the distant background star. Approximating the background star to be infinitely far away, simple geometry gives the distance to the star as $r = 1AU/\theta$. The angle θ is called the parallax of the star. Stars up to 300 light years away, ~~can be~~ their distance can be determined by this method.

The H R diagrame



e location of main sequence stars in the
eoratical H-R diagrame. The points give
e theoratical position of stars which have
t begun their lives as main sequence stars
(zero-age main sequence)

The Hertzsprung-Russel Diagrame

Many properties of stars, even their distance can be determined in terms of the location of a star in the H-R diagrame.

The H-R diagrame is a plot of the Luminosity (L) of a star versus its effective temperature (Te). Lets first see how L and Te are determined.

Luminosity (L)

To obtain the luminosity of a star we need to make two measurements.

(a) We need to measure the apparent brightness 'f' of a star, which is the total energy received from the star/unit time/unit area at the Earth.

(b). We need the distance r of the star away from us. For nearby stars and star clusters distance can be calculated by parallax method or moving cluster method.

Then $L = f\, 4\pi r^2$ //

Effective Temperature

There are two independent methods for measuring Te. They are

(1) UBV color Photometry

The baisic idea behind this is to measure the propotions of radiant energy put out by a thermal body at ultraviolet (U)

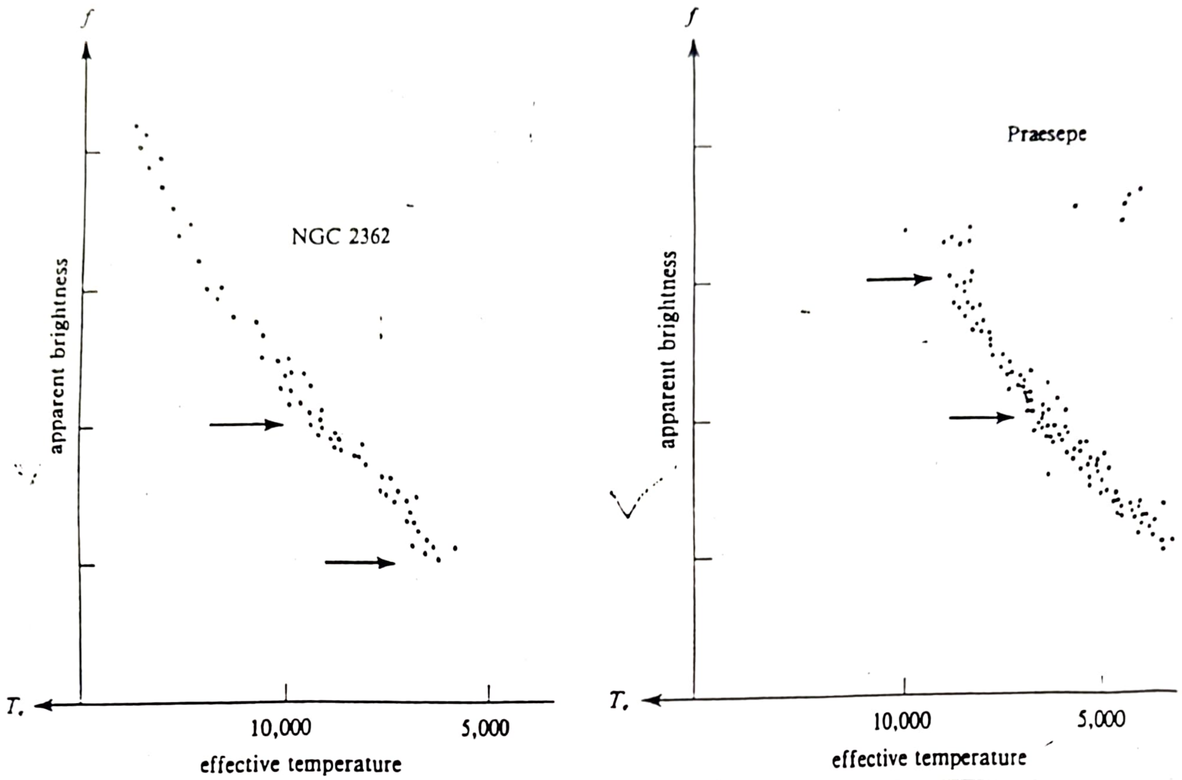


Figure 9.7. The H-R diagrams of the NGC 2362 and the Praesepe open clusters. The segments of main-sequence stars between the two arrows in each diagram refer to intrinsically similar stars: the segment in Praesepe appears displaced upward from the segment in NGC 2362 only because the former cluster is closer

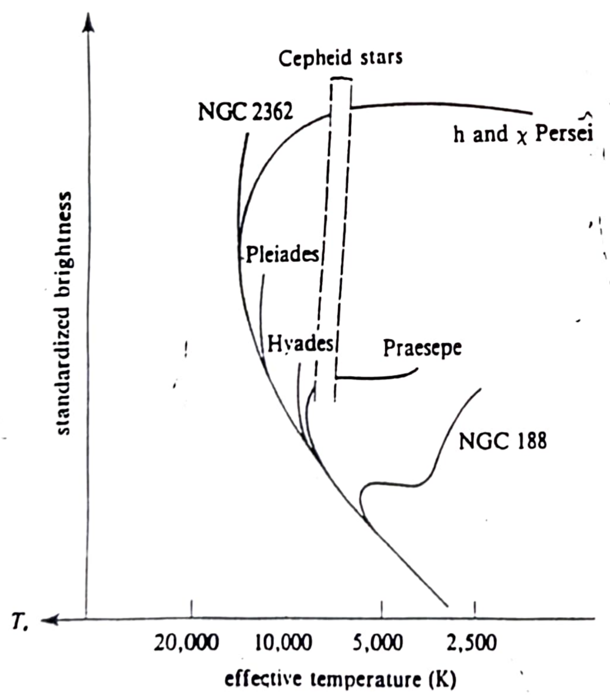


Figure 9.8. Schematic composite H-R diagram for open clusters whose main sequences have been "lined up" with the Hyades. Stars are conventionally not plotted in the strip indicated by dots, because stars whose mean state falls in this part of the diagram ("Cepheid instability strip") invariably prove to be pulsating stars. In addition, a gap (the Hertzsprung gap) usually exists between the turnoff point on the main sequence and the red giants, because stars evolve so quickly through this region that there is little probability of seeing them there. (Adapted from Allan Sandage, *Ap. J.*, 125, 1957, 435.)

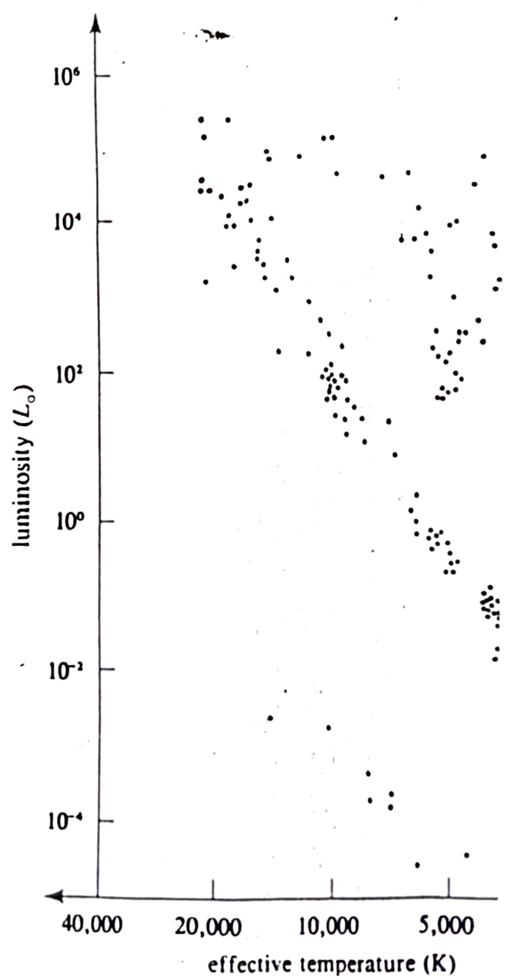


Figure 9.5. The Hertzsprung-Russell diagram of ne
Main-sequence stars and white dwarfs are fai
delineated.

Blue (B), and visual (V) wavelengths. These proportions depend on the surface temperature of the opaque body; the hotter the body, greater is the ~~body~~ proportion of shorter wavelengths.

If $f(U)$, $f(B)$ and $f(V)$ are the apparent brightness of a star at U, B and V wavelengths respectively, then

$$f(V)/f(B) = \text{function of } T_e$$
$$f(B)/f(U) = \text{function of } T_e$$

② Spectral classification

The basic idea behind spectral classification is that for a given chemical composition, the pattern of absorption lines which are formed in the photosphere of a star depend on the temperatures which exist in the photosphere. Thus to a first approximation, the spectral type of a star yields an estimate of its effective temperature.

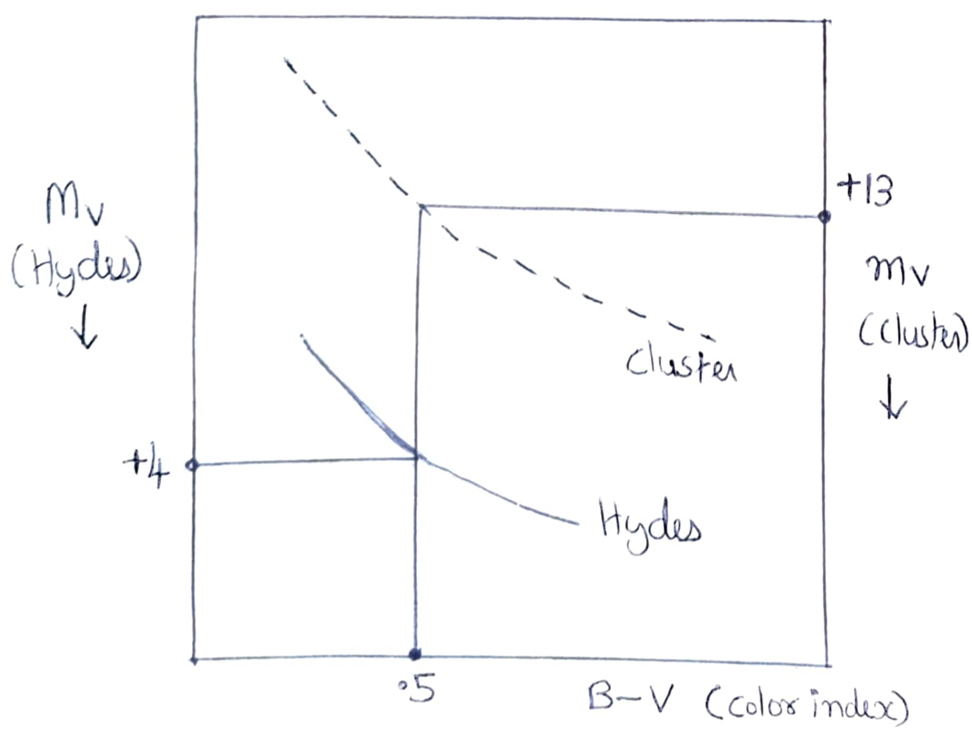
$$\text{Spectral type} = \text{function of } T_e$$

b) The H.R Diagramme of Star Clusters and Main Sequence Fitting (H.R diagramme used in determing distances)

Observationally astronomers have found two different kinds of star clusters in the galaxy, ie the Open clusters and the globular clusters. A clusters angular

Main Sequence Fitting

Theoratical models show that all unevolved stars of the same chemical composition have the same locus in the colour-magnitude diagrame: The Zero Age Main Sequence (ZAMS). If we assume that all clusters have the same chemical composition then their cluster colour-magnitude diagrames should be the same. For most clusters we observe only (m_V, B-V), and obtain the shape of the main sequence. For the Hydes we know (M_V, B-V), and we can determine the distance modulus $m_V - M_V$ for each cluster by fitting its ZAMS to that of the Hydes (See Fig below). The vertical shift needed at a given B-V to fit the two ZAMS's together over their region of overlap is just the distance modulus of the cluster.



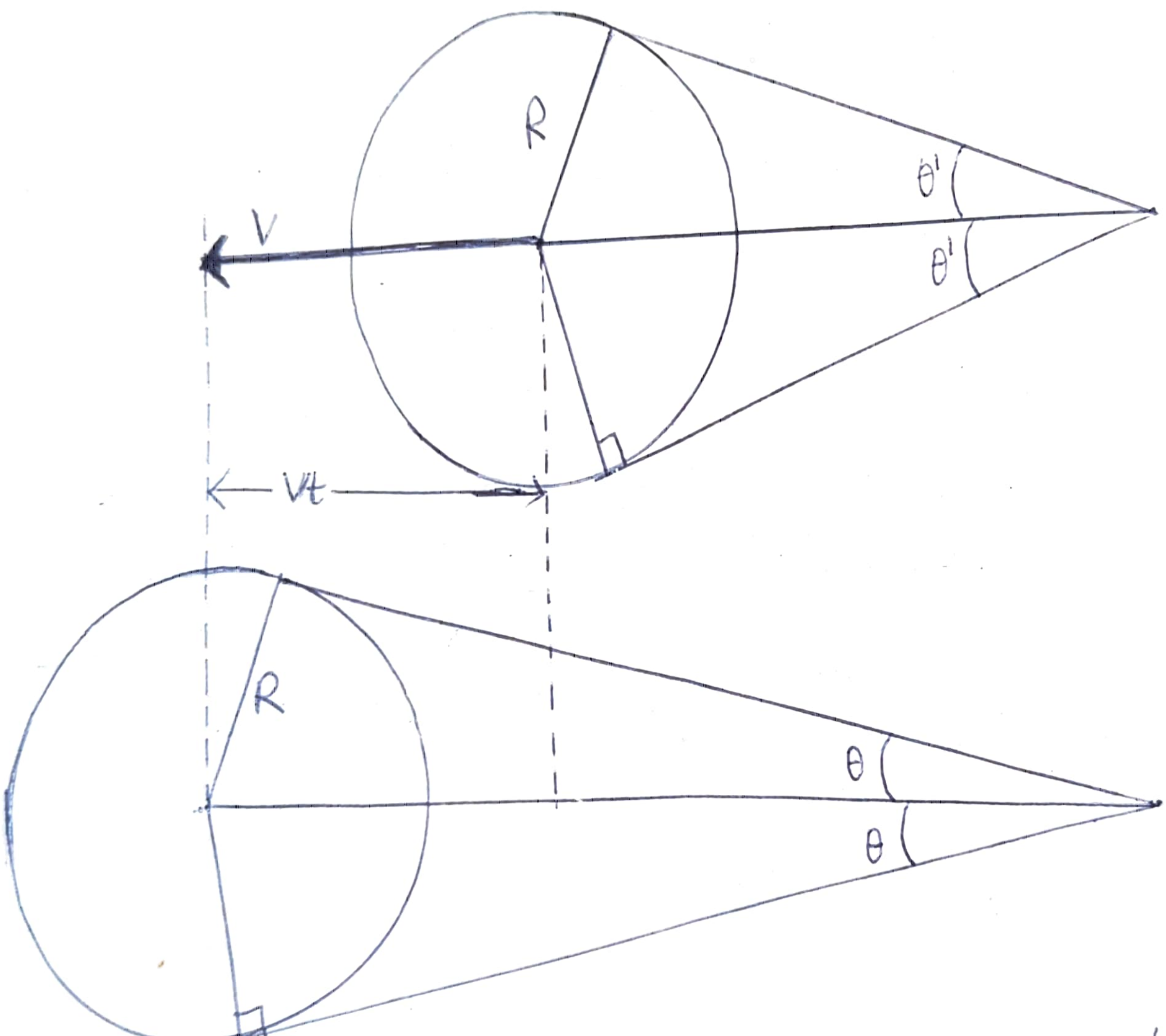
The difference between the apparent (m_V) and absolute (M_V) magnitudes at a given colour is independent of color and is the distance modulus of the cluster. (in this case it is 13-4 = 9) size is ~~at~~ always small, so all stars can be approximated to lie at the same distance from us.

H-R diagrame of Open clusters (Determination of their distances)

The H-R diagrame of two open clusters NGC 2362 and Paraesepe are plotted in fig (9.7). The arrows in the figure mark of segments in NGC 2362 and Paraesepe which contain the same range of effective temperatures, and in which the stars presumably comprise the same range of masses. The fact that the black arrows are higher in Praesepe than those in NGC 2362 must then mean that the Praesipe cluster is closer to us so that the same stars appear brighter.

Trumpler was the first to compare the relative distance between clusters by means of the observational H-R diagrame. (It must be noted that in the observation H-R diagrame apparent brightness f replaces the luminosity L. This is because f is measurable while L is not. The form of the diagrame is not changed by this replacement.) He did this by sliding the observational H-R diagrame up or down until the main sequence line up. The amount that we have to slide the diagrame tells us how much further or nearer the cluster is relative

The Moving Cluster Method



The Hydes star cluster has a component of velocity v directed away from the observer. At an earlier time the angular size appears to be $2\theta'$. After a time t, the angular size appears to be smaller 2θ. The change the angular size must ~~yeild~~ be due to increased distance size ... yeilds the relation to some standard. Fig (9.8) shows the result for a number of open clusters when the Hydes cluster is taken as standard. If we can deduce the distance to the Hydes cluster, we can deduce the distance to all other open clusters on the diagramme from the amount that we have to slide them up to make their main sequence line up with that of Hydes.

Distance to the Hydes Cluster - Moving Cluster Method

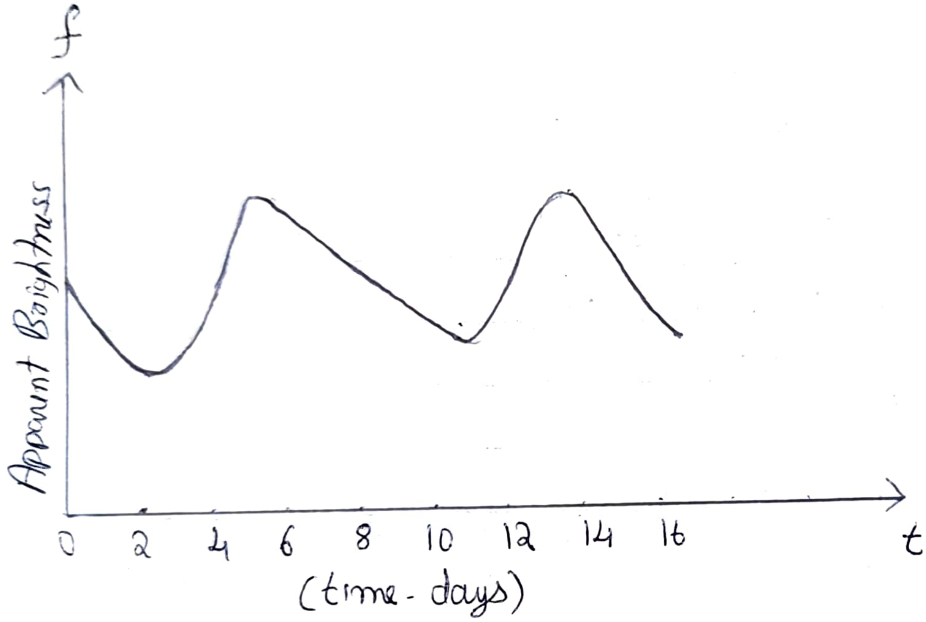
The Hydes cluster happen to be close enough to us and happens to be moving with a component of velocity directed away from our line of sight that it has a deducible change of angular size with time. These facts allow the distance to Hydes to be calculated by what is called "the moving cluster method". The geometric ideas are simple and is shown in fig (9.10)

For star clusters that are too far away to allow astronomers to measure the properties of their relatively faint main-sequence stars, there is another way to discover their distances by the use of cephid variable stars.

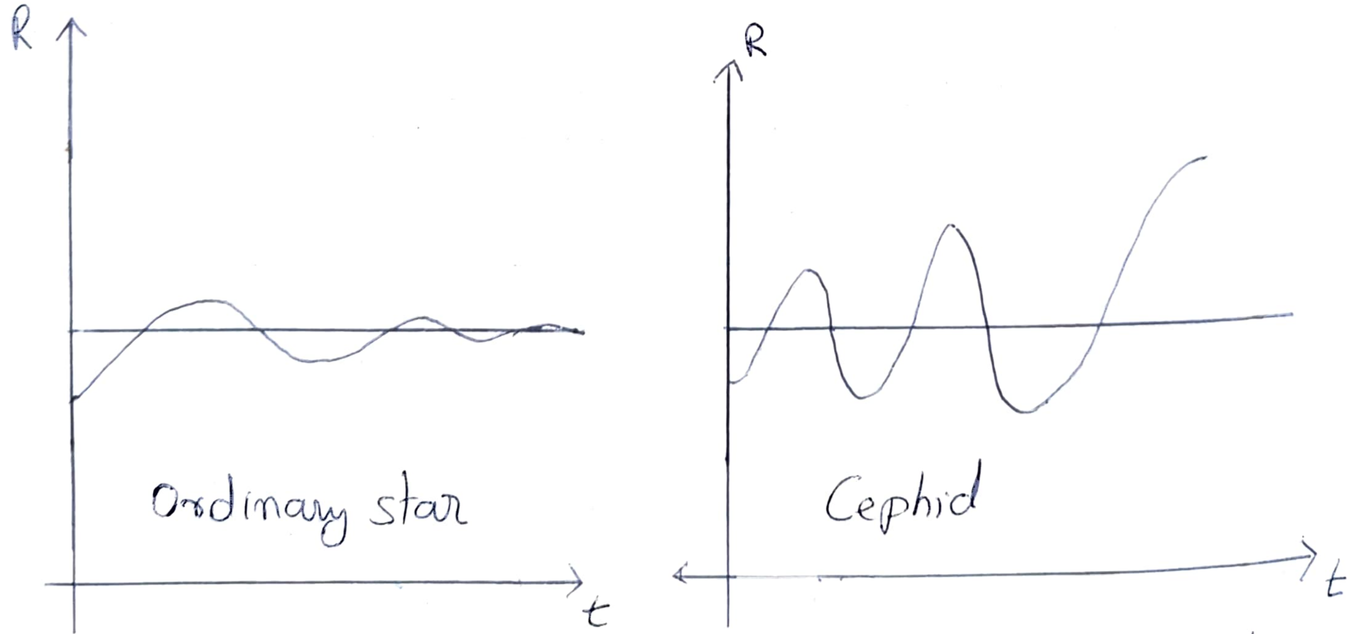
Cephid's as Distance Indicators

In many star clusters are found stars which vary in total light output ~~in~~

Cephids



The light curve of a Cephid Variable.
The period is here about a week.



Radius - time relation of a star and Cephid.

(luminosity) in a periodic fashion. See fig
Such stars are called variable stars the most famous of them being the stars called cephids. The luminosity (L) of the cephids are directly proportional to their pulsation period - (P) ie $L \propto P$

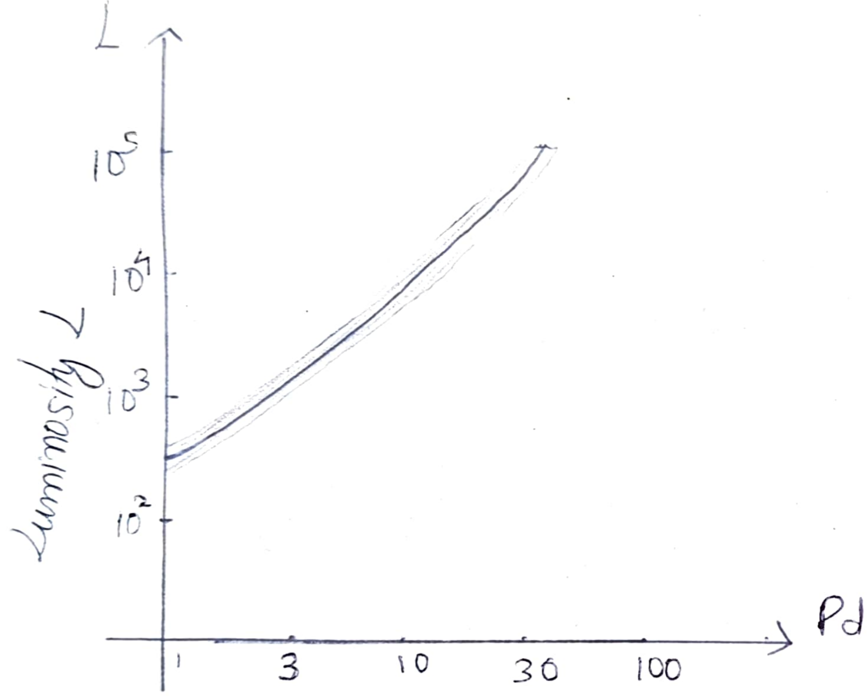
But we have $L = f 4\pi r^2$ where f is the apparent brightness. and r the distance of the star from us. Since the apparent brightness of the cephids can be easily determined we can calculate their distance from us by the relation $L = KP = f 4\pi r^2$; where K is a constant of proportionality.

There are a small number of open clusters that contain cephids and we can use main-sequence fitting method to find their distances. With this data, the cephid period luminosity relationship can be calibrated and the constant of proportionality found out. The P-L relation of cephids provided the major method by which astronomers extended the distance scale beyond the nearby open clusters.

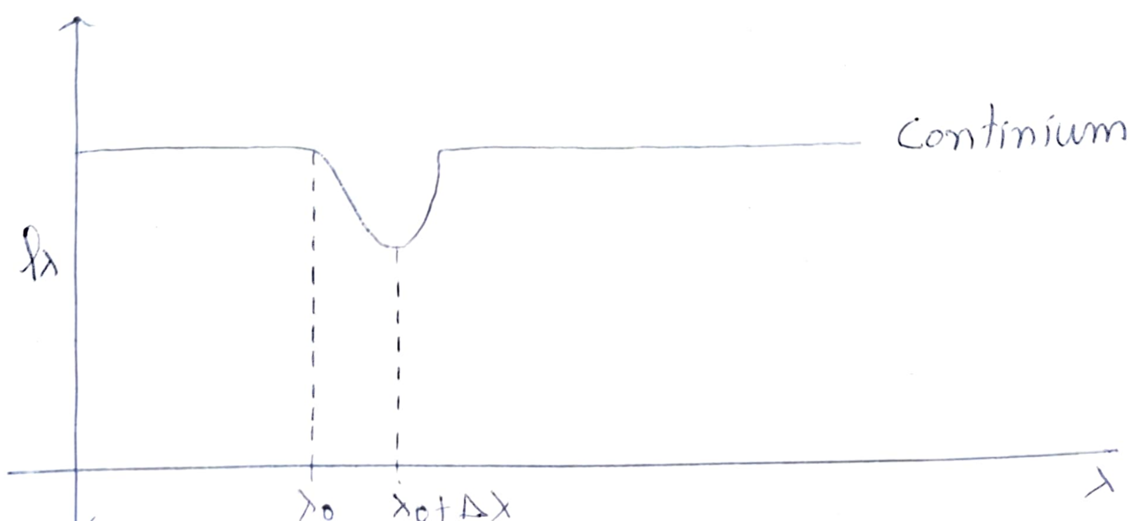
3 Expansion of the Universe

Our present understanding of the nature of galaxies started with the work of Hubble in 1920's. He was the first to discover cephid variables in Andromeda galaxy M31 and

The Period-Luminosity Relationship for Classical Cepheid



An Absorption line Spectrum of a Galaxy



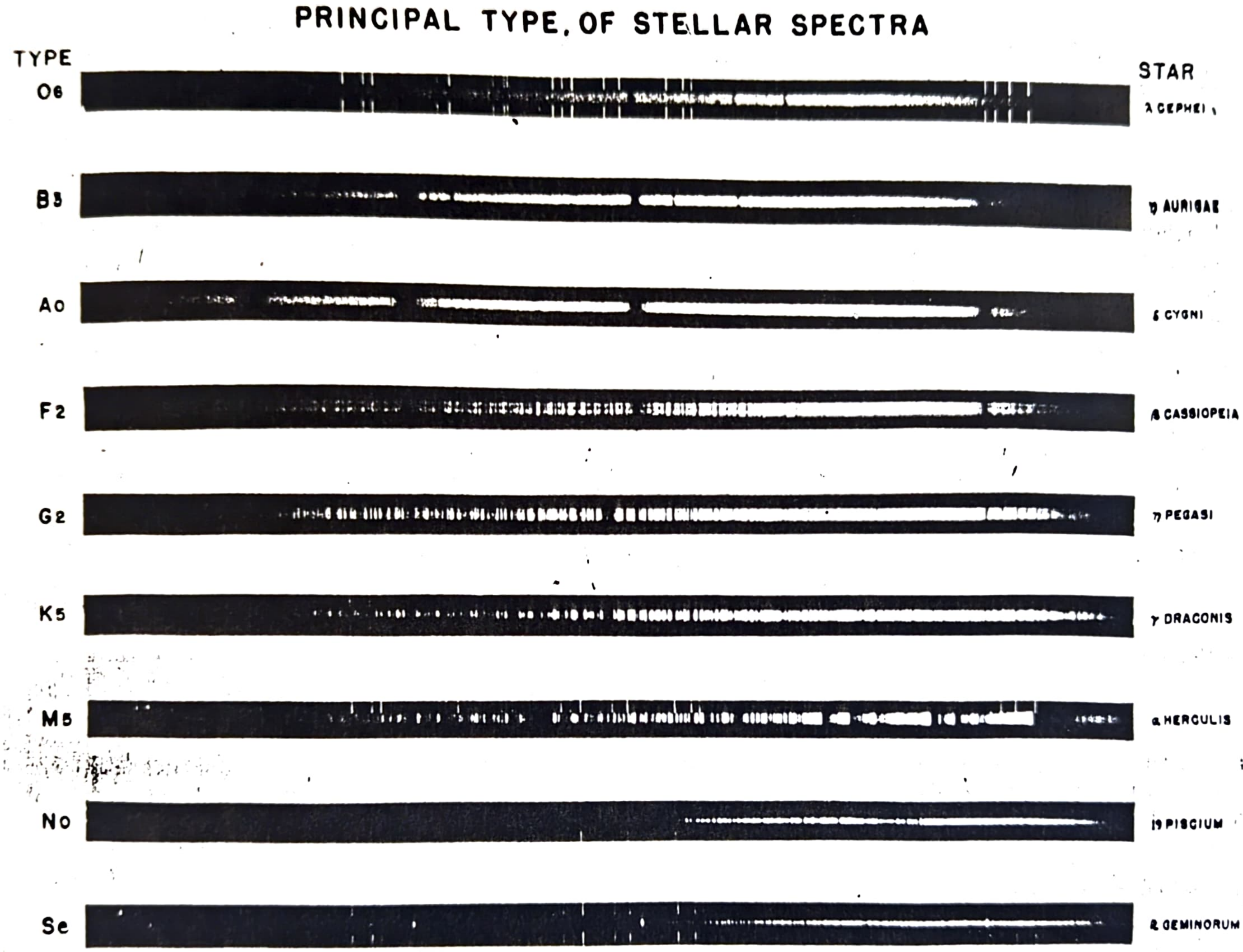
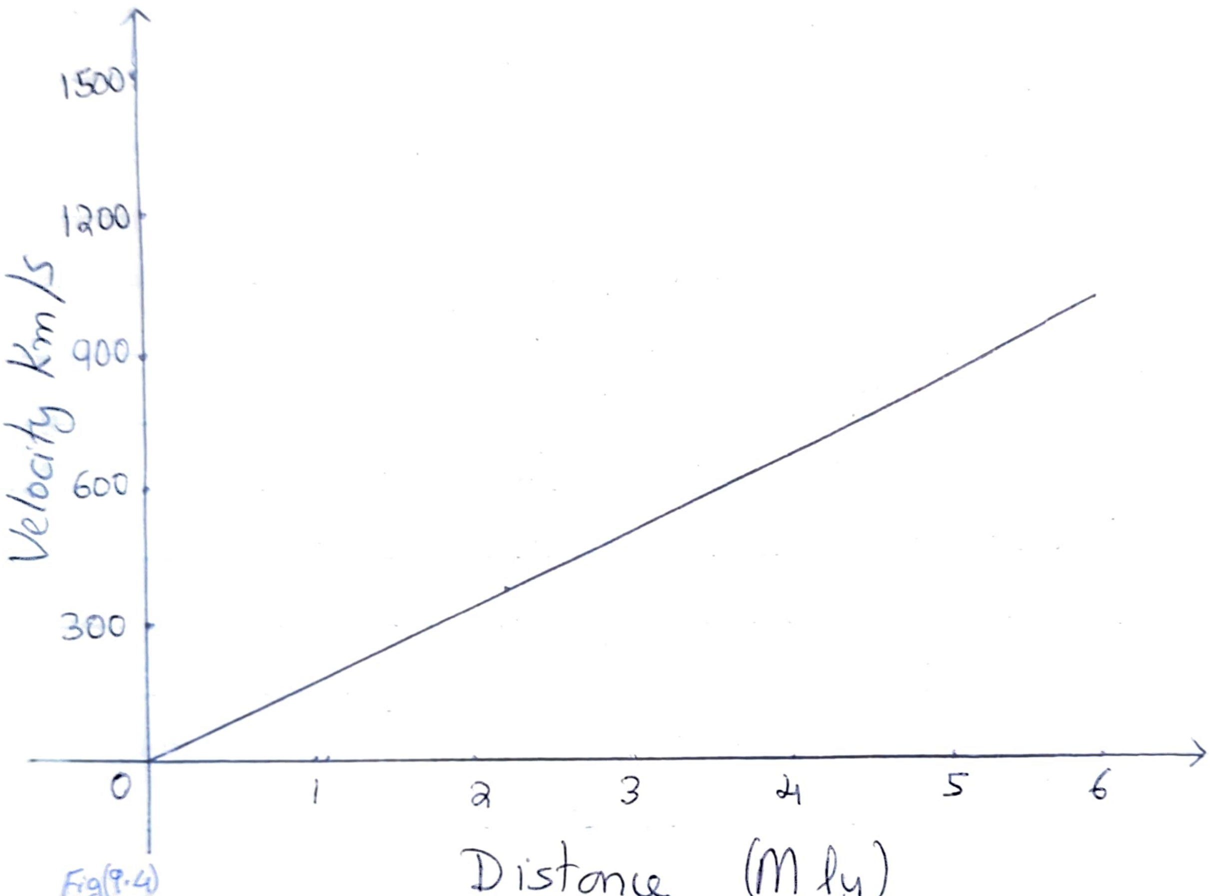


Figure 9.3. Photographic examples of stellar spectra. The star's spectrum is in the middle of each strip; the bright lines above and below each stellar spectrum yield reference wavelengths from a laboratory source. Notice that the hydrogen absorption lines (dark stripes against a bright background) are strongest in the A0 star, whereas the "e" attached to the Se star indicates that it exhibits emission lines (bright stripes against a less bright background in addition to the more common absorption lines. (Palomar Observatory, California Institute of Technology.)

Hubble's Law



Fig(9.4)

cluster) were receding fastest at 1200 km/s. Hubble plotted a graph with distance against redshift velocities of galaxies and he got a straight line. See fig (1.4). That is redshift velocity was directly proportional to the distance of the galaxy. This relation between velocity (V) and distance R, $V = H_0R$ is called the "Hubbles law" and the constant of proportionality is called the Hubbles constant.

b The implications of Hubble's law

Hubble had been reluctant to draw such a grand conclusion from his flimsy data. For one the redshifts might not be classic Doppler shifts at all. Moreover there were perhaps 100 billion galaxies in the sky and so far the expansion was based on less than 200 of them. Nevertheless the Hubble's law was widely accepted. In 1933 Humason had measured a cluster of galaxies in constellation Ursa Major. The Hubbles law still held at the distance.

b1 The age of the Universe

The expansion of the Universe has an implication that at some time in the past, all the matter in the universe must have been extremely closely packed. This thought was the seed of the big bang theory and

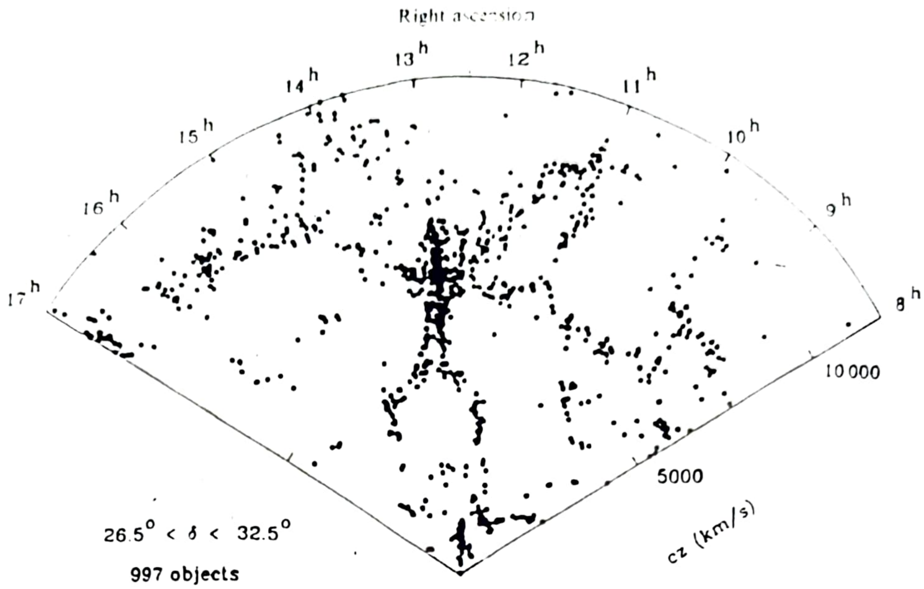


Fig. 13.4. Filaments and voids in the galaxy distribution. The body of the 'dancing man', and his arms and legs, represent long chains of galaxies, and the spaces around the man are largely empty of galaxies. This is a two-dimensional slice through the local neighbourhood, with distance measured radially (in km s^{-1}, assuming the Hubble relation) and angle on the sky as the other coordinate. (Reprinted by permission from *The Astrophysical Journal*, 1991.)

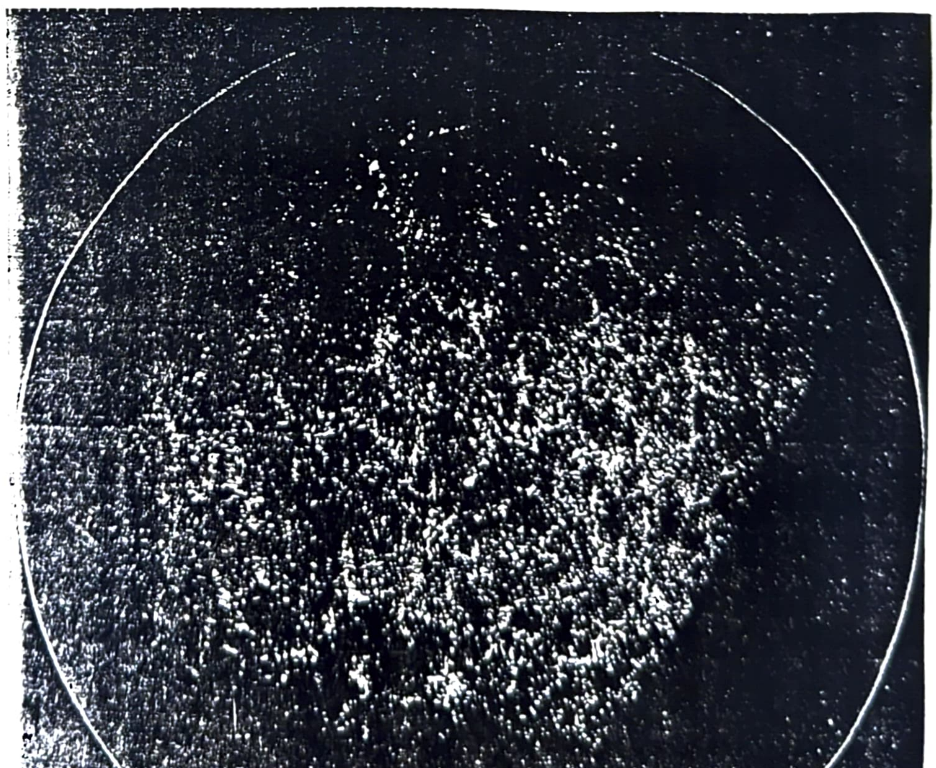


Fig. 13.3. The Lick catalogue of one million galaxies, represented as a map of the sky. The sharp edge to the distribution shows where the survey is cut off by obscuration in the plane of the Galaxy. (Reprinted by permission from *The Astronomical Journal*, 1977.)

that the Universe had a beginning in time. The distance from us (R) that any given galaxy has travelled during its life time (t_0) is equal to velocity times life time.

ie $R = V t_0$. Now by Hubbles law $V = H_0 R$

ie $R = H_0 R t_0$ or $t_0 = \frac{1}{H_0}$ //. That is the quantity $1/H_0$ is a measure of the "expansion age of the Universe".

<u>Determination of the large scale structure of the Universe</u>

Because of the simple linear relationship between velocity and distance and the ease with which redshifts can be measured, redshift is now used as a measure of distance for galaxies at large distance ($\gtrsim 20$ Mpc). With the advent of fibre optic spectrographs more than a hundred galaxy redshifts can be measured in a single exposure. This has transformed our view of the Universe, allowing the three dimensional structure of the Universe on a large scale to be probed for the first time.

Olber's Paradox

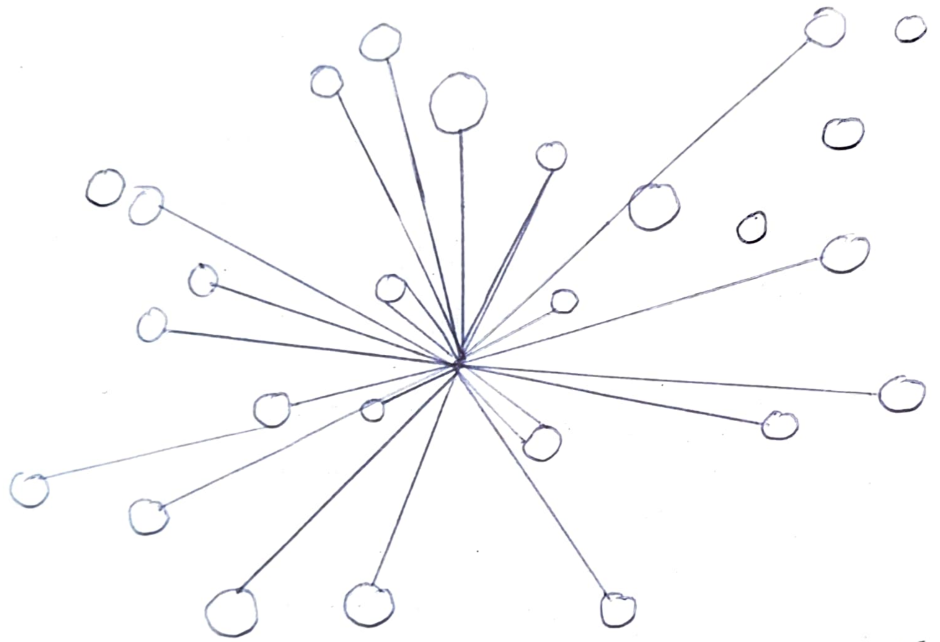


Fig-1

Resolution of Obler's Paradox

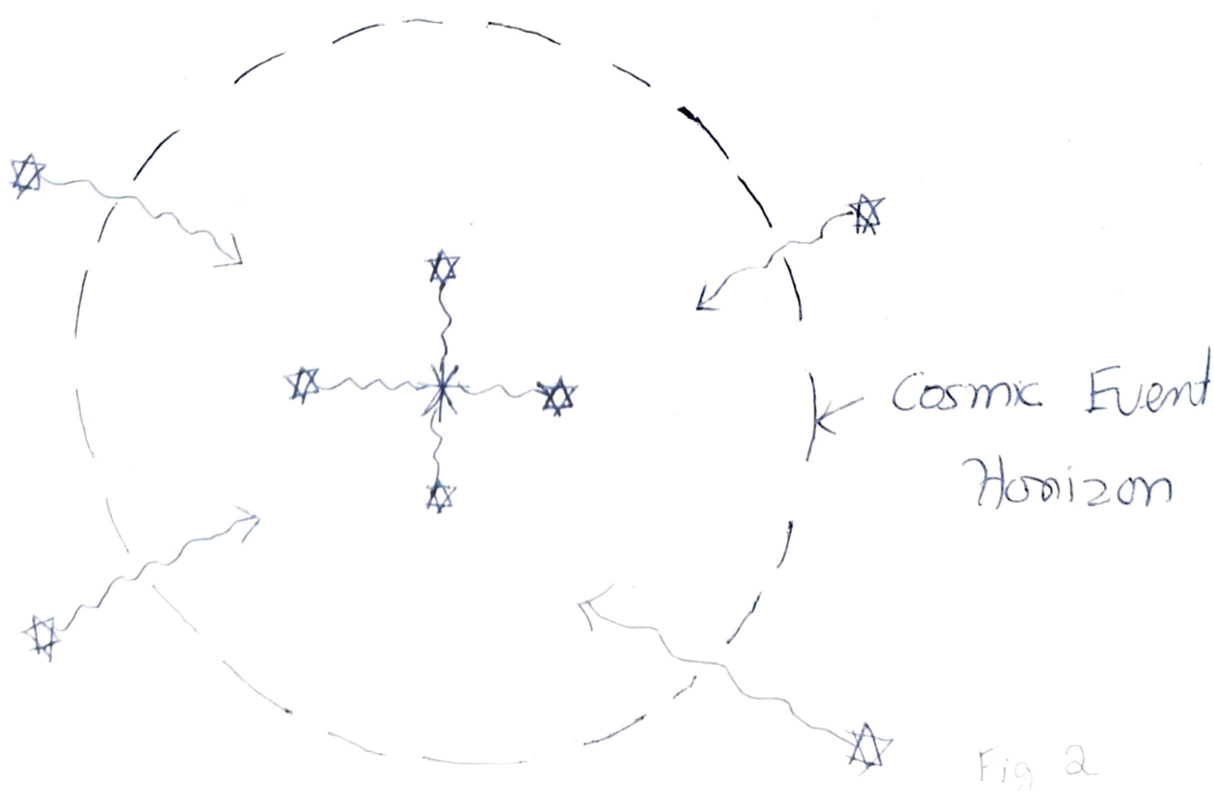


Fig 2

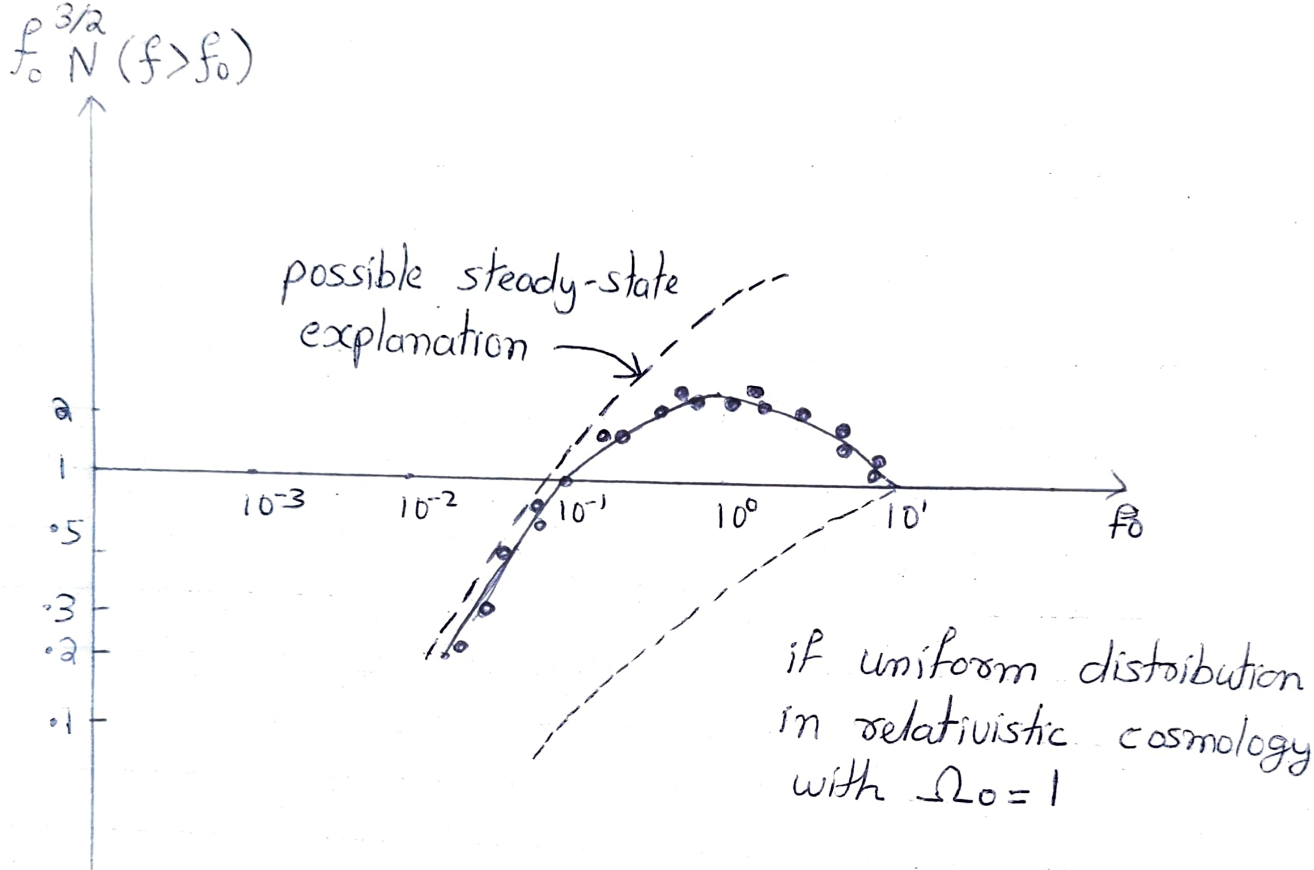
The Big Bang Model of the Origin of the Universe

1 **Distribution of Radio galaxies and Quasars**

The first piece of evidence in favour of Big bang model concerned spatial distribution of strong radiogalaxies. Counts of these sources by Ryle and his collegues revealed that there were more radio galaxies at far distances from us than at near distances. Now when we look at very distant objects, we are looking at events from the remote past. The excess of strong radiogalaxies in the remote past must mean that the conditions in the past were different from what they are today. An excess of far sources would be a global property of the universe and would therefore contradict the basic tenet of the steady-state theory

2 **Olbers Paradox and its resolution**

In a Euclidean Universe uniformally and statically filled with stars on average, every line of sight will eventually intercept the surface of a star and the night sky would be as bright as the surface of an average star see fig (1). (This has an analogy to being in the midst of a forest and every where around you, you see a wall of trees).



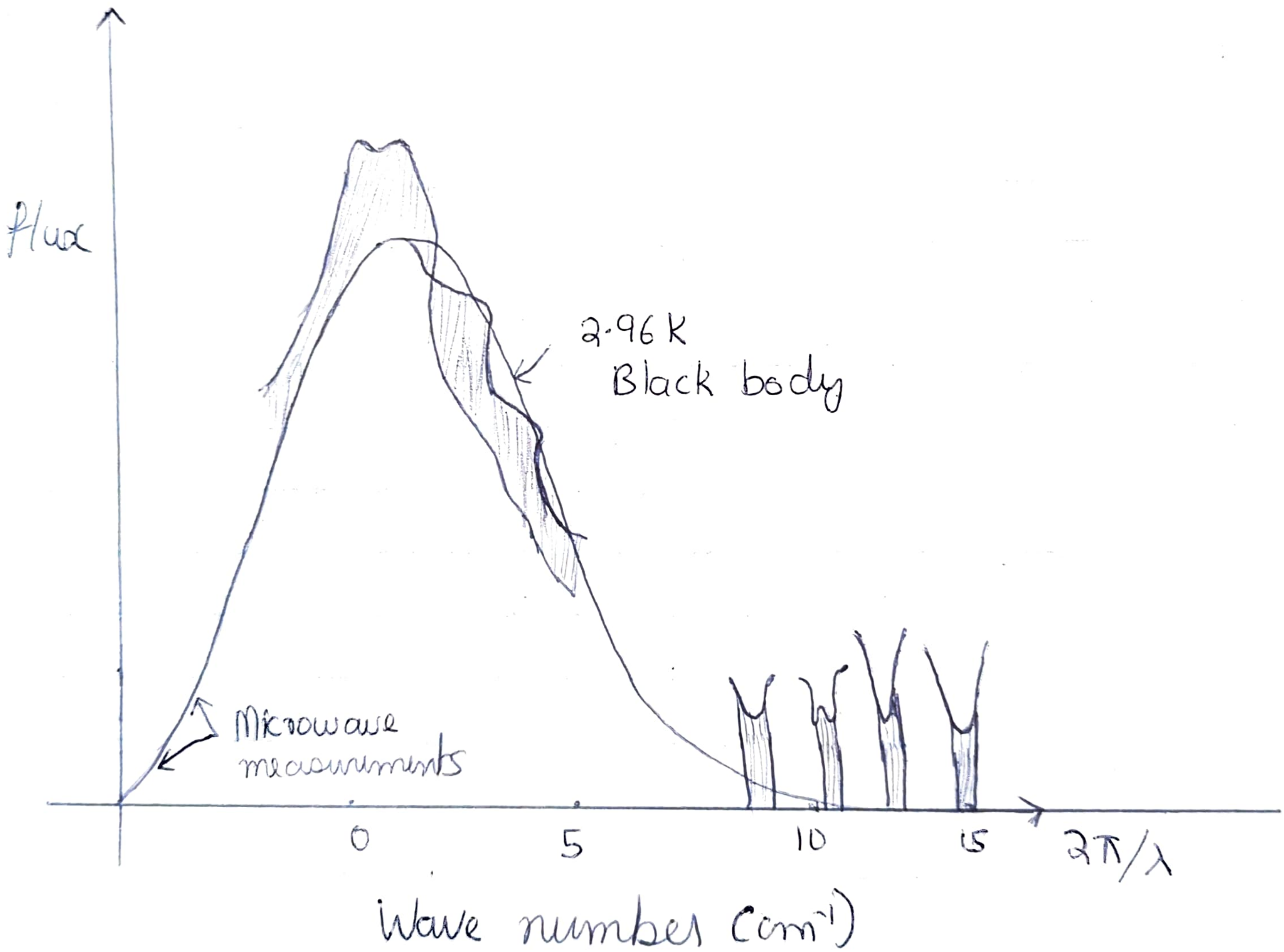
$f_0^{3/2} N(f>f_0)$
2
1
·5
·3
·2
·1
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
f_0
possible steady-state explanation
if uniform distribution in relativistic cosmology with $\Omega_0 = 1$

As E.R Harrison has emphasized, in conventional cosmologies there are logically two independent ways ~~of~~ out of this dilemma ie (1) expansion of the Universe and (2) a finite age for the Universe. Expansion helps because the cosmological redshift prevents distant stars from making simple Euclidean contribution to the night-sky brightness. But most important is the assumption of the finite age of the Universe, because the light from very distant stars then simply does not have time to get here, even if we ignore redshift (see fig (2)) In the steady state only the expansion argument is available. In Big-bang theories cosmological expansion and a finite age of the Universe go hand in hand.

The Prediction of Cosmic Background radiation

Around 1950 Gamov proposed that many of the heavy elements currently found in the Universe were synthesized by thermonuclear reactions when the universe was very young. To enable such reactions to proceed fast enough to offset the rapid decrease of density caused by the expansion of the early universe, the matter in the Big bang must have started at very high temperatures

Spectral Energy Distribution of the Cosmic - Microwave Background Radiation



The bracketed points show some of the early radio measurements. The shaded region between the solid curves show the measurements of Woody and Richards.

and associated with it would be a thermal radiation feild whose energy density greatly exceeded that of matter. As the universe expanded both matter and radiation would cool, become too low for further nuclear reactions and eventually the matter and radiation will follow seperate thermal histories. Past this epoch the radiation is no longer in thermodynamic equilibrium with matter but it would continue to cool as each photon is redshifted by the cosmological expansion. Under these conditions it can be shown that the radiation continues to maintain a thermal (ie black body) spectral distribution characterized by lower and lower radiation temperatures as the Universe continues to expand. According to early estimates of Gamow, by now the radiation would be no more than tens of degree above absolute zero.

Dickie and Peebles worked to improve the Big bang nucleosynthesis calculations when it become clear that stellar nucleosynthesis could probably not explain the present cosmic abundance of He. In particular they concluded that the best

Figure 15.1. Deep galaxy map of a square section of the sky, 6 on a side, made from data supplied by Rudnicki and colleagues at the Jagellonian University in Cracow. The average galaxy among the more than 10,000 counted may be located between two and three billion light-years away in this map. Notice that the distribution on this scale is almost random, in contrast with Figure 14.6. (For details, see Groth, Peebles, Seldner, and Soneira, *Scientific American*, **237**, May 1977, 76.)

Figure 16.5. Arno Penzias (right) and Robert Wilson standing in front of the horn antenna which was used to detect the cosmic microwave background in 1965. (Bell Labs photo)

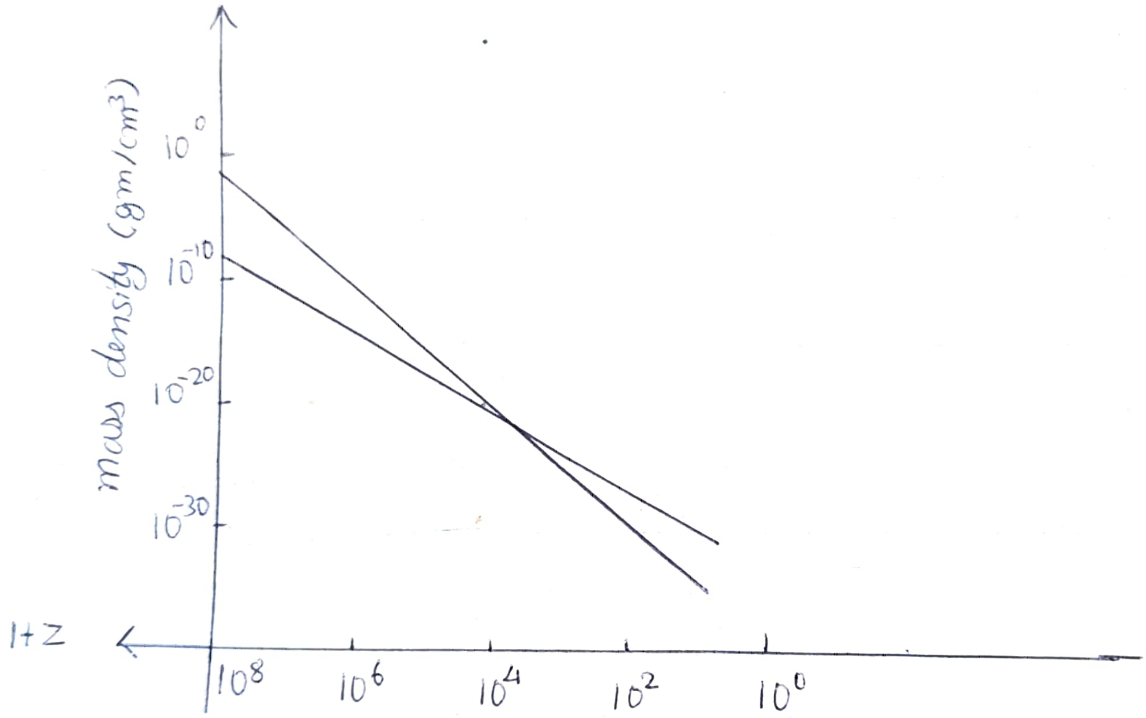
available data led to a prediction of a few Kelvin of the present radiation temperature. They set out to build an appropriate antenna to detect this radiation.

Discovery of Cosmic Microwave Background Radiation

It was at this time that two Bell Lab scientist Arno Penzias and Robert Wilson was calibrating a small radiohorn designed for satellite communication. They became deeply puzzled by the ubiquitous presence of excess noise in their sources even after they have eliminated all noise sources. This seemed to be distributed throughout the sky in an isotropic fashion. When they consulted Bernand Bruke of MIT of this problem, Bruke realized that Penzias and Wilson had probably found the cosmic background radiation that Dickie and his collegues were planning to measure. Put into touch with each other the two groups subsequently published simultaneous papers detailing the prediction and discovery of a universal thermal radiation feild with a temperature of about 3K.

The discovery of cosmic microwave background radiation was a direct prediction of the Big-bang theory and effectively sounded the death knell for the steady state

Evolution Of Mass Density of Radiation ρ_{rad} and Ordinary matter ρ_m as function of expansion of the Universe.



To obtain a timescale requires adopting a specific cosmological model, and this depends on how we treat the neutrino contribution.

viewpoint.

The Hot Big Bang

We shall now discuss the theoratical implications of Penzias-Wilson discovery. Since the wavelength λ_{max} of peak thermal emission varies with the radiation temperature T_{rad} in accordance with Wiens displacement law, $\lambda_{max} \times T_{rad} =$ a constant

and since this peak wavelength must strech with the expansion of the Universe in accordance with the redshift formulae, $\lambda_{max} \propto D(t)$, where $D(t)$ is the scale factor of the Universe, then the radiation temperature T_{rad} must decrease as the Universe expands according to

$$T_{rad} \propto D^{-1}$$

Thus in the past when D was smaller the density of the thermal radiation feild must have been larger

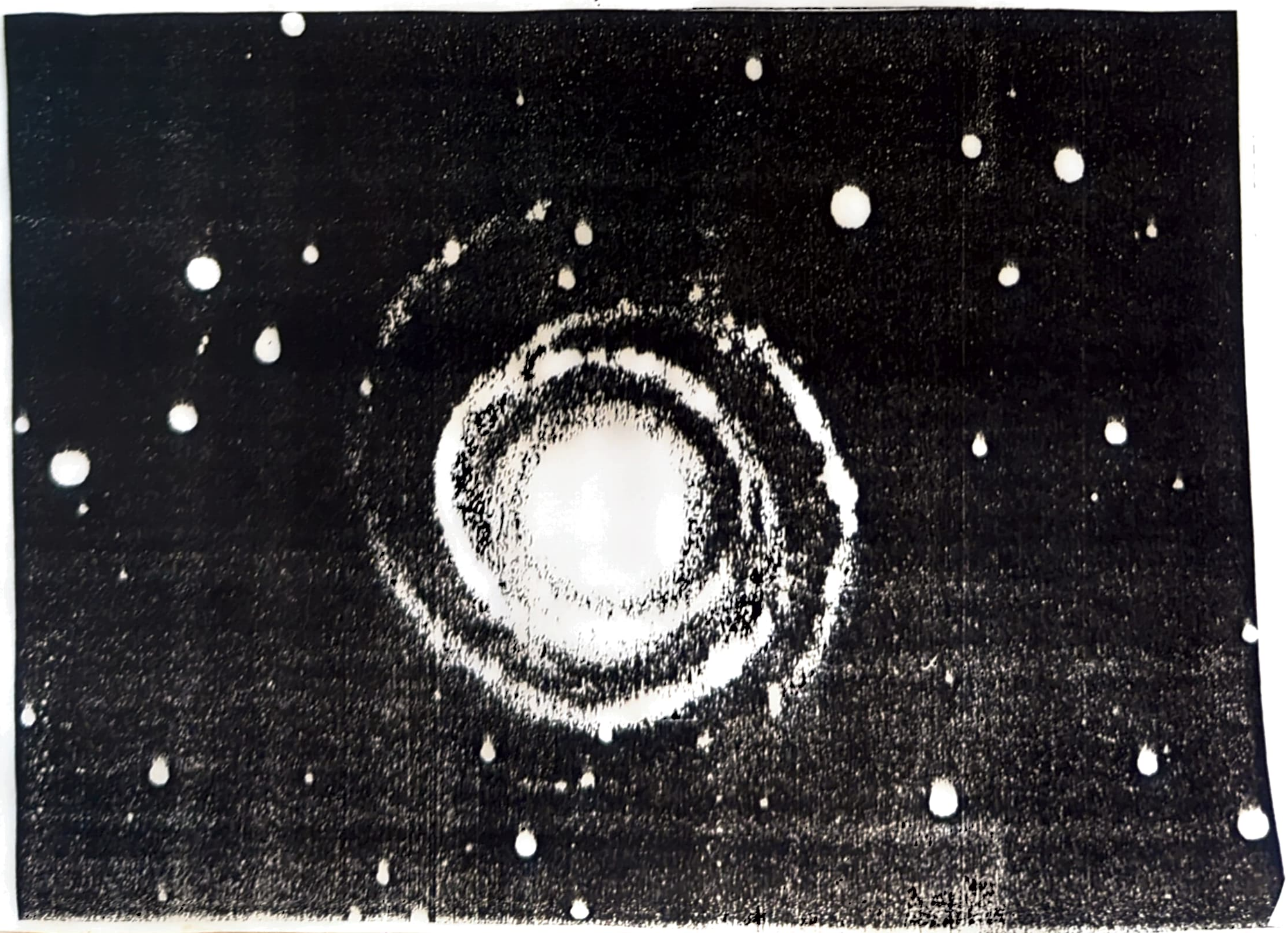
$$\rho_{rad} \propto D^{-4}$$

In contrast the matter density in the past would have been higher only according to $\rho_m \propto D^{-3}$. The ratio ρ_{rad}/ρ_m can be written in the from $\frac{\rho_{rad}}{\rho_m} \propto D^{-1}$

The current ratio approximately equals 10^{-3} if we include only the matter seen in

Figure 12.23. The face-on spiral galaxy NGC 4622 (type Sb) may resemble the Milky Way System if we could imagine placing ourselves near the top of the picture in the spur that joins the two outer spiral arms. Notice the "beads on a string" appearance of the optical spiral arms, and the fact that strong dust lanes can be readily seen on the *inside* edges of the optical arms. (The Cerro Tololo Inter-American Observatory.)

Figure 9.6. (a) The open clusters h and χ Persei, and (b) the globular cluster M3. The angular diameters of h and χ Persei are about half a degree each, and the angular diameter of the bright core of M3 is about 3 minutes of arc. The small angular extent of a star cluster implies that all the stars belonging to the cluster lie at nearly the same distance from us. (Lick Observatory photograph.)



galaxies to calculate ρ_{m_0}. Thus at present matter dominates radiation. But at a redshift of about 10^3, and the universe was a 1000 times smaller than it is now, the mass density of matter and radiation would have been equal. Before that the Universe would have been radiation dominated.

Summarizing in words the above mathematical deductions we have; If we go backwards in time, the radiation density and the matter density both increase, but the radiation density increases faster than matter density. Thus in the beginning radiation might have dominated over the ordinary forms of matter. The early universe must have been exceedingly bright. That is according to the Big Bang theory the universe had a beginning and it began with all the matter of the Universe in a primoordial high-density state which exploded in ~~the~~ a big bang. From that instant on the Universe expanded. As the galaxies formed they too shared in the expansion.

Big Bang Nucleosynthesis

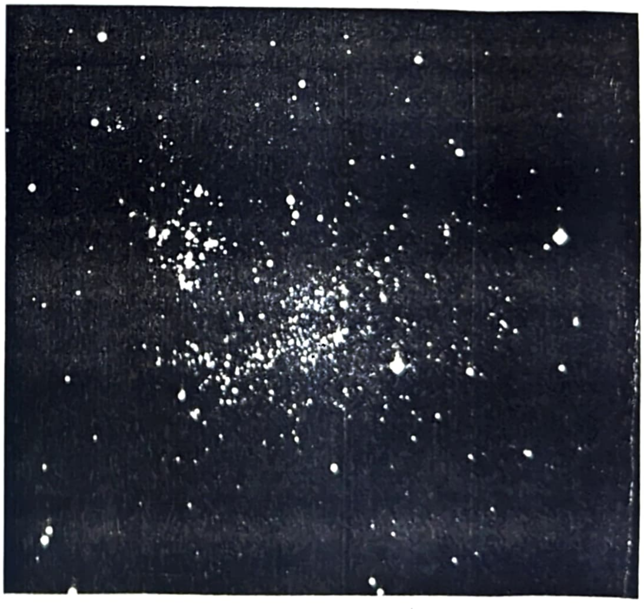
(The abundance - Confirmation of Big Bang theory)

At a temperature in excess of 10^{12} K, (t less than 10^{-4} s) the creation and

(a)

(b)

(c)



(d)

Figure 13.6. Examples of (a) an elliptical galaxy, (b) an ordinary spiral galaxy, (c) a barred spiral galaxy, and (d) an irregular galaxy. (From A. Sandage, *The Hubble Atlas of Galaxies*, Carnegie Institution of Washington, 1961; photographic materials from Palomar Observatory, California Institute of Technology.)

annihilation of baryon, antibaryon pair could have been in thermodynamic equilibrium with ambient radiation field. Because of the slight asymmetry of baryons and antibaryons, there would be a residue of P_r proton and neutron left without partners to annihilate when temperature drops significantly below 10^{12} K. The residual baryons are kept in thermal equilibrium by the presence of enormous sea of elementary particles via the reactions

$$\left.\begin{array}{l} {}^{1}_{1}P + {}^{0}_{-1}e \rightleftharpoons {}^{1}_{0}n + {}^{0}_{0}\nu \\ {}^{1}_{1}P + {}^{0}_{0}\bar{\nu} \rightleftharpoons {}^{1}_{0}n + {}^{0}_{+1}e \end{array}\right\} \rightarrow ①$$

When the universe is tens of seconds old, the neutrinos and antineutrinos will have decoupled from electrons and positrons. The absence of e^- and $\bar{\nu}$ with sufficient energy to drive reaction ① in the forward direction means that the free neutrons will want to beta decay.

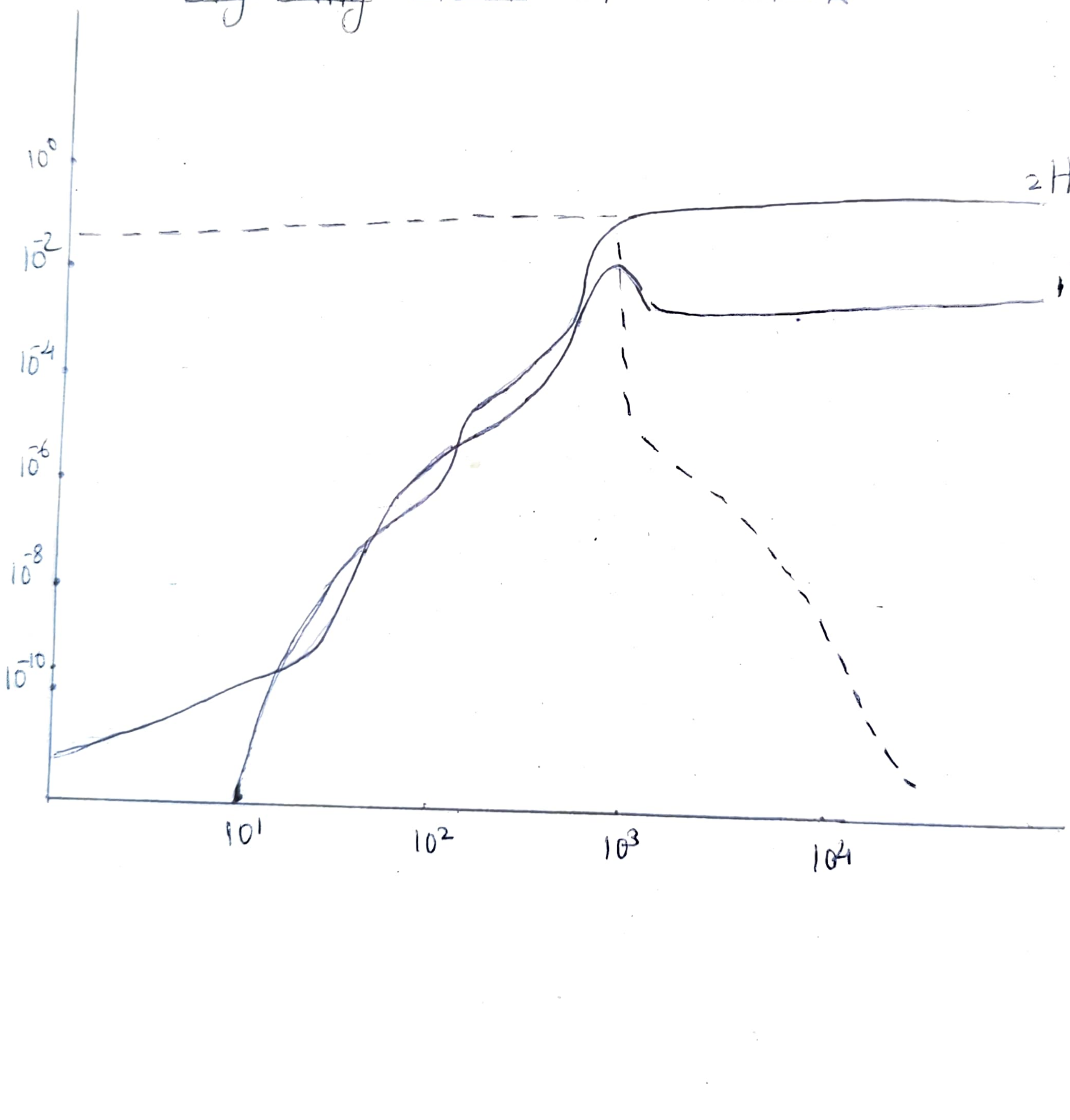
$${}^{1}_{0}n \rightarrow {}^{1}_{1}P + {}^{0}_{-1}e + {}^{0}_{0}\bar{\nu}$$

However well before 10 minutes is up the free neutrons may be captured by free protons to from deuterium nucleus

$${}^{1}_{0}n + {}^{1}_{1}P \rightarrow {}^{2}_{1}H + {}^{0}_{0}\gamma$$

Once deuterium has formed, a rapid sequence of reaction leads mostly to the

Big Bang Models of WAGONER



formation of He-4. Two neutrons and two protons are needed to from an He-4 nucleus.

Now the energy required to make a neutron from a proton must baisically be derived from an energetic electron or antineutrino. As long as such particles are around the relative concentration of neutrons and protons will satisfy the relation

$$\frac{N_n}{N_p} = \exp\left[-(m_n - m_p)c^2 / KT\right]$$

As an application of the equation we can see that there would be roughly one neutron for every five protons at a temp. $T = 10^{10}$ K.

As already stated two neutrons and two protons are needed to from one He-4 nucleus. Thus if we started with 2 neutrons and 10 protons (1n for every 5p) we would end with one He-4 nucleus and 8 H nuclei (protons). The atomic weight of He-4 is 4, whereas we have a total atomic weight 2+10=12. Thus this simple calculation yeilds an expected fraction of He-4 by mass of 4/12 = 33%. More sophisticated analysis yield a He mass fraction close to 25%. Observationally stars are believed to begin their lives with a He

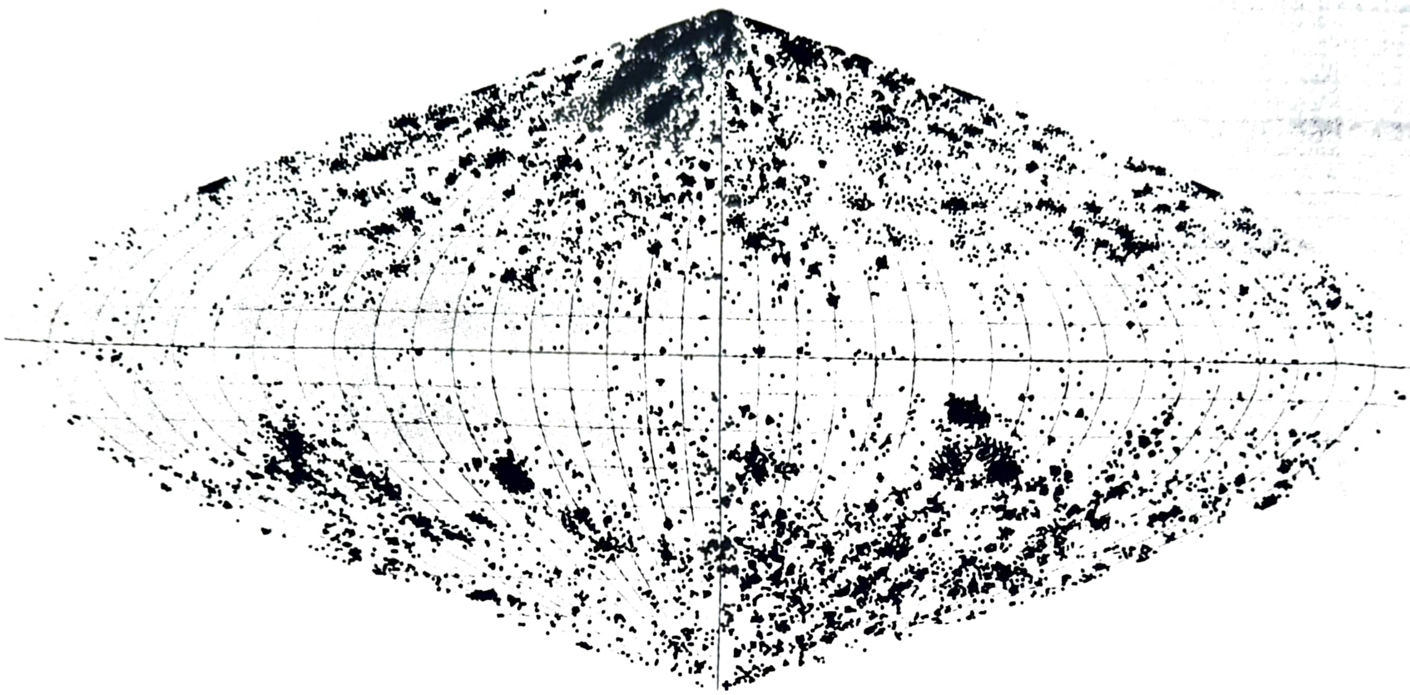


Figure 13.3 A map of the distribution in Galactic latitude and longitude of NGC objects made by Charlier in the 1920s. The abbreviation NGC stands for *New General Catologue*, which was published by John Dreyer in 1895 as a revision of John Herschel's *General Catalogue of Nebulae*. Notice that relatively few spirals can be found near the Galactic equator. (Figure kindly supplied by Paul Gorenstein.)

Figure 13.4. An edge-on spiral, NGC4565 (type Sb). Notice the characteristic dark belt of absorbing material in the principal plane of the disk. (From A. Sandage, *The Hubble Atlas of Galaxies*, Carnegie Institution of Washington, 1961; photographic material from Palomar Observatory, California Institute of Technology.)

mass fraction of 20% to 30%. The reasonable agreement of this range with that predicted by standard Big-Bang models is cause for optimism that we are on the right track.

Conclusion

Man can only imagine between the extremes of the tiniest and the vast. But the timescale and distance scale that he comes across in his scientific search are either too tiny or too vast for man to imagine or decipher it. And hence he fails to understand the truth ingrained in the heart of the atom and that same truth embedded over the stars. It has been just 10,000 years since the earliest neolithic man began to settle and farm in earnest, to the the present. And what great advancements have we made in that short time span. In the begining of the 19^{th} century scientist were confident that by the end of the century they would be able to explain most phenomena. Far from it, mysterious phenomena like missing mass and large scale structure of the universe has been unveiled with no current theories to account for it. The more we learn, the more we find there is to learn. It may be another 10,000 years before questions like What is matter? What is light, What is gravity etc are answered in their full sense. To those men of the future we may seem like neolithic men of science beginning to settle near the banks of the starry eternal rivers of truth.

Cowley

Lauve

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